

3. Functions of discrete random variables

Let X and Y be independent r.v.

$$X: \begin{pmatrix} x_1 & x_2 & x_3 \\ 0 & 1 & 2 \\ 0,2 & 0,5 & 0,3 \end{pmatrix}, \quad Y: \begin{pmatrix} y_1 & y_2 \\ -1 & 0 \\ 0,3 & 0,4 \end{pmatrix}$$

Det PMF of $2X + Y^2$ and $2X \cdot Y^2$

$$Y^2: \begin{pmatrix} 0 & 1 \\ 0,4 & 0,3 \end{pmatrix} \rightarrow \text{we switched the numbers to keep the values in increasing order}$$

$$P(Y^2=0) = P(Y=0) = 0,4$$

$$P(Y^2=1) = P(Y=-1) = 0,3$$

$$2X: \begin{pmatrix} 0 & 2 & 4 \\ 0,2 & 0,5 & 0,3 \end{pmatrix}$$

$$P(2X=0) = P(X=0) = 0,2$$

$$P(2X=2) = P(X=1) = 0,5$$

$$P(2X=4) = P(X=2) = 0,3$$

$$2X + Y^2: \begin{pmatrix} 0+0 & 0+1 & 2+0 & 2+1 & 4+0 & 4+1 \\ 0 & 1 & 2 & 3 & 4 & 5 \\ 0,14 & 0,06 & \dots \dots \end{pmatrix}$$

$$P(2X + Y^2 = 0) = P(2X=0 \cap Y^2=0) = P(2X=0) \cdot P(Y^2=0) = 0,2 \cdot 0,4 = 0,14$$

$$P(2X + Y^2 = 1) = P(2X=0 \cap Y^2=1) = P(2X=0) \cdot P(Y^2=1) = 0,2 \cdot 0,3 = 0,06$$

At the end check if the sum of the probabilities is equal to 1 !!

$$2X \cdot Y^2: \begin{pmatrix} \begin{matrix} 0 \cdot 0 \\ 0 \cdot 2 \\ 0 \cdot 4 \end{matrix} & \begin{matrix} 1 \cdot 2 \\ 1 \cdot 4 \end{matrix} & \begin{matrix} 2 \cdot 2 \\ 2 \cdot 4 \end{matrix} \\ \begin{matrix} 0 \\ 2 \\ 4 \end{matrix} & \begin{matrix} 0,15 & 0,09 \\ 1 - (0,15 + 0,09) \end{matrix} \end{pmatrix}$$

$$\begin{aligned} P(2X \cdot Y^2 = 0) &= P[(2X=0 \cap Y^2=0) \cup (2X=0 \cap Y^2=1) \cup \\ &\quad (2X=2 \cap Y^2=0) \cup (2X=4 \cap Y^2=0)] = \\ &= P(2X=0) \cdot P(Y^2=0) + P(2X=0) \cdot P(Y^2=1) + \\ &\quad P(2X=2) \cdot P(Y^2=0) + P(2X=4) \cdot P(Y^2=0) \end{aligned}$$

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

4. Averages

Expected Value and Variance of a Discrete R.V.

Def: A mode of a discrete r.v. X is a number, denoted x_{mod} such that the probability $P_X(x_{\text{mod}}) \geq P_X(x)$, $\forall x \in S_X$

A median of a discrete r.v. X is a number, denoted x_{med} such that the probability $P_X(X > x_{\text{med}}) = P(X < x_{\text{med}})$

The expected value of X is the number

$$E(X) = \sum_{x \in S_X} x \cdot P_X(x)$$

Prop:

$$1. E(aX+b) = a E(X) + b, \quad \forall a, b \in \mathbb{R}$$

$$2. E(X+Y) = E(X) + E(Y), \quad \forall X, Y \text{ - r.v.}$$

$$3. E(X - E(X)) = 0$$

Ex: $X: \begin{pmatrix} -1 & 0 & 1 \\ 0,2 & 0,3 & 0,5 \end{pmatrix}$

$$\underline{E(X)} = -1 \cdot P_X(-1) + 0 \cdot P_X(0) + 1 \cdot P_X(1) =$$

$$= -1 \cdot 0,2 + 0 \cdot 0,3 + 1 \cdot 0,5 = \boxed{0,3}$$

Remark!

x_{mod} and x_{med} are not unique!!

Ex: For one quiz, 10 students had the following grades:

9; 5; 10; 8; 4; 7; 5; 5; 2; 4;

$x_{\text{mod}} = ?$ $x_{\text{med}} = ?$ these grades are smaller than 7

$x_{\text{mod}} = 5$ (arise more often)

$x_{\text{med}} = 7$

why 7? because the number of grades that are smaller than 7 is equal with the number of grades that are bigger than 7

Def: The variance of X is the number denoted by $\text{Var}(X) = E[(X - E(X))^2]$ ($\text{var}(x) = D^2(x)$ in romanian)

The standard deviation of X is the number denoted by $\sigma(X) = \sqrt{\text{Var}(X)}$

Properties:

1. $\text{var}(X) \geq 0$, $\forall X$ - discrete r.v.

2. $\text{var}(aX + b) = a^2 \cdot \text{var}(X)$

3. $\text{var}(a) = 0$, $\forall a \in \mathbb{R}$

4. $\text{var}(X) = E(X^2) - [E(X)]^2$

Ex: $\text{var}(X) = E(X^2) - [E(X)]^2 = 0,7 - 0,3^2 = 0,7 - 0,09 = 0,61$

$$E(X^2) = \sum_{x \in S_X} x^2 P_X(x) = (-1)^2 \cdot 0,2 + 0^2 \cdot 0,3 + 1^2 \cdot 0,5 = 0,7$$

$$\sigma(X) = \sqrt{0,61}$$



Def: The n^{th} moment of X is $E(X^n)$

The n^{th} central moment of X is $E[X - E(X)]^n$

$\text{var}(X) = 2^{\text{nd}}$ central moment of X

Remark!

The "normal values of X " are in

$$[E(X) - \sigma(X), E(X) + \sigma(X)]$$

6. Conditional PMF of a r.v. X

X - discrete r.v. $\rightarrow$ PMF $P_X(x)$

B - event, $P(B) > 0$

Def: The conditional PMF of X given the occurrence of B :

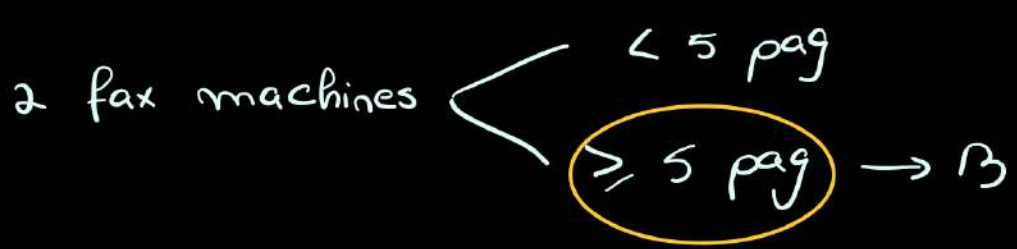
$$P_{X|B}(x) = P(X=x|B)$$

Prop:

$$P_{X|B}(x) = \begin{cases} \frac{P_X(x)}{P(B)}, & x \in B \\ \text{zero}, & \text{otherwise} \end{cases}$$

Ex: The length of a fax is a discrete r.v.

$$P_X(x) = \begin{cases} 0,15, & x = 1; 2; 3; 4 \\ 0,1, & x = 5; 6; 7; 8 \\ \text{zero}, & \text{otherwise} \end{cases}$$



a. PMF of the second machine

b. $E(X|B)$, $\text{var}(X|B)$, $\sigma(X|B) = ?$

$$B = \{5; 6; 7; 8\} \Rightarrow P(B) = 4 \cdot 0,1 = 0,4$$

$$P_{X|B}(x) = \begin{cases} \frac{P_X(x)}{P(B)}, & x \in B \\ \text{zero, otherwise} \end{cases} = \begin{cases} \frac{0,1}{0,4}, & x \in \{5; 6; 7; 8\} \\ \text{zero, otherwise} \end{cases} =$$

$$= \begin{cases} \frac{1}{4}, & x = 5; 6; 7; 8 \\ \text{zero, otherwise} \end{cases} = \begin{pmatrix} 5 & 6 & 7 & 8 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

$$\text{var}(X|B) = E(X^2|B) - [E(X|B)]^2$$

$$P_Y(x) = 5^2 \cdot \frac{1}{4} + 6^2 \cdot \frac{1}{4} + 7^2 \cdot \frac{1}{4} + 8^2 \cdot \frac{1}{4} - \left(\frac{13}{2}\right)^2 = \dots = a$$

$$\sigma(X|B) = \sqrt{a}$$

Ex: X - the number of additional years that a person (T_0) will live

if the person has high blood pressure, noted with (H), then X has a geometric with $p = 0,1$. Otherwise, if the person has low blood pressure, noted with (R), then X has a geometric with $p = 0,05$.

a. $P_{X|H}(x), P_{X|R}(x) = ?$

b. 40% of all persons have high blood pressure, find PMF of X

$$\Rightarrow P(H) = \frac{40}{100} = \frac{2}{5} \Rightarrow P(R) = 1 - \frac{2}{5} = \frac{3}{5}$$

a. $\left(\text{gem: } P_X(x) = \begin{cases} p(1-p)^{x-1}, & x = 1, 2, \dots \\ \text{zero, otherwise} \end{cases} \right)$

$$P_{X|H}(x) = \begin{cases} 0,1 \cdot 0,9^{x-1}, & x = 1, 2, \dots \\ \text{zero, otherwise} \end{cases}$$

$$P_{X|R}(x) = \begin{cases} 0,05 \cdot 0,95^{x-1}, & x = 1, 2, \dots \\ \text{zero, otherwise} \end{cases}$$

conditional
PMFs

b. $\{H, R\}$ - space event

Law of Total Probabilities

$$\begin{aligned} P_X(x) &= P(H) \cdot P_{X|H}(x) + P(R) \cdot P_{X|R}(x) = \\ &= \begin{cases} \frac{2}{5} \cdot 0,1 \cdot 0,9^{x-1} + \frac{3}{5} \cdot 0,05 \cdot 0,95^{x-1}, & x = 1, 2, \dots \\ \text{zero, otherwise} \end{cases} \end{aligned}$$

Ex: N - number of packets $\rightarrow$ depends on whether the page has graphic images

if the page has image (i) $\Rightarrow N$ is uniformly distributed between 1 and 50 packets

if the page has just text (T) $\Rightarrow N$ is uniformly distributed between 1 and 5 packets

A page has images with probability $\frac{1}{4}$

a. $P_{N|i}(x), P_{N|T}(x) = ?$

b. $P_N(x) = ?$

c. $P_{N|N \leq 3}(x) = ?$

d. $E(N | N \leq 3) = ?$

e. $\text{var}(N | N \leq 3) = ?$

$$a. P_{N|i}(x) = \begin{cases} \frac{1}{50}, & x=1, 2, \dots, 50 \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$P_{N|T}(x) = \begin{cases} \frac{1}{5}, & x=1, 2, 3, 4, 5 \\ \text{zero}, & \text{otherwise} \end{cases}$$

b. $\{i, T\}$ - space event

$$P_N(x) = \begin{cases} \overset{\uparrow \frac{1}{4}}{P(i)} \cdot P_{N|i}(x) + \overset{\uparrow \frac{3}{4}}{P(T)} \cdot P_{N|T}(x), & x=1, 2, 3, 4, 5 \\ P(i) \cdot P_{N|i}(x), & x=6, 7, 8, \dots, 50 \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{4} \cdot \frac{1}{50} + \frac{3}{4} \cdot \frac{1}{5}, & x \in \{1 \dots 5\} \\ \frac{1}{4} \cdot \frac{1}{50} = \frac{1}{200}, & x \in \{6 \dots 50\} \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$c. P_{N|N \leq 3}(x) = \begin{cases} \frac{P_N(x)}{P(N \leq 3)}, & x \in \{1, 2, 3\} \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$P(N \leq 3) = P(N=1) + P(N=2) + P(N=3) = 3 \cdot \frac{31}{200}$$

$$\Rightarrow P_{N|N \leq 3} = \begin{cases} \frac{\frac{31}{200}}{3 \cdot \frac{31}{200}} = \frac{1}{3}, & x \in \{1, 2, 3\} \\ \text{zero}, & \text{otherwise} \end{cases}$$

$$d. E(N | N \leq 3) = 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} = \frac{6}{3} = 2$$

$$e. \text{var}(N | N \leq 3) = E(N^2 | N \leq 3) - [E(N | N \leq 3)]^2 = \\ = 1^2 \cdot \frac{1}{3} + 2^2 \cdot \frac{1}{3} + 3^2 \cdot \frac{1}{3} - 2^2 = \frac{2}{3}$$